

Characterisation and PPO-assisted Optimisation of Nonlinear Dynamics in a Memristive Transneuron Device

F230992. A. N

Department of Physics, Loughborough University, LE11 3TU, United Kingdom

Abstract: The projected demands on computing are set to outpace conventional hardware. A promising alternative is neuromorphic computing, but configuring individual neuromorphic devices for systems level integration remains an unsolved challenge. Through numerical continuation analysis, a deterministic reduction for the transneuron device developed by Midya et al. was characterised. We discuss the presence of five distinct regimes, exhibiting notable features such as tristability, independently controllable bursting timescales, and an empirical model for firing rate. A surrogate assisted PPO framework was developed and demonstrated successful convergence to a target selectivity configuration with an explained variance of 0.997, establishing proof of concept for automated co-design and optimisation of neuromorphic hardware.

Introduction

For half a century, information technology (IT) has followed a consistent trend. Known commonly as Moore's law, it posited that the number of transistors within a single microchip would double every two or so years [1]. A less frequently considered implication of this law was its predicted 25% annual decline in energy consumption per processing unit [2]. Whilst it has proven to be a robust prediction, its loosening grasp upon the integrated circuit (IC) industry has been foreseen and observed for some time [3, 4, 5]. Meanwhile, global computational demands show no signs of slowing to accommodate this shift in progression [6, 7]. Thus, research into new and alternative methods of computation has become a pertinent focus of the computing research community [8, 9].

One such alternative, neuromorphic computing, approaches information processing from a fundamentally different perspective. Inspired by biological methods of computation, neuromorphic architectures differ from the current paradigm through the co-localisation of processing and memory [10]. This eliminates the slow and costly constant transfer of data between separate physical components, frequently referred to as the von Neumann bottleneck, that dominates the power budgets of conventional hardware [11]. Biological systems demonstrate that this approach can achieve remarkable performance with regards to speed, latency, robustness, and reliability [12, 13]. Perhaps the most remarkable example is the brain, capable of complex perception, motor control, and reasoning. These systems also exhibit significant plasticity and adaptability, enabling online learning through dynamic and local reconfiguration [14, 15]. This has the potential to mitigate the costly process of back propagation required for most conventional applications of machine learning. A fundamental step towards the development of neuromorphic hardware lies in understanding, adapting, and emulating these systems. Furthermore, the field is active and growing rapidly, attracting attention and investment across both the academic and commercial research communities [16, 17].

When trying to understand something complex, an enduring strategy has been to reduce the system until left with a single universal unit. This approach has been formalised within neuroscience as the neuron doctrine [18]. Under this pretext, it follows that to design neuromorphic systems capable of complex computation, we must first have a firm grasp on the individual components

from which the whole is composed. At the device level, memristors have emerged as one of the most compelling building blocks for neuromorphic hardware. Originally derived axiomatically by L. Chua [19] and finally experimentally realised by HP Labs [20], these two-terminal devices naturally emulate key properties of biological synapses and neurons through their current history dependent non-linear dynamics [21, 22].

Currently, the characterisation of memristive synapses is more advanced, with well-established metrics and broad materials coverage [23, 24, 25]. While neuronal emulators are progressing, understanding is limited and less unified, often framed as an open challenge. A notable recent development is the demonstration of the transneuron by Midya et al. [26]. The device proposed is a single volatile diffusive memristive circuit capable of reconfiguring its spiking dynamics. By Leveraging various dynamical regimes, the transneuron is able to emulate the behaviour of multiple functionally distinct neurons found in separate parts of the brain. However, the promise of such devices can only be realised if they can be reliably and reproducibly configured to a desired operational state, a challenge that the field has yet to systematically address.

The neuromorphic research community currently operates at two largely disconnected scales of integration, device physics and systems architecture [27, 28]. At the device level, research focuses on material characterisation, switching dynamics, and stochastic behaviour. At the systems level, research focuses on network architecture, routing, and collective computation. Currently missing is the translation layer between the two, a mechanism for mapping desired computational behaviour to optimal fabricable device parameters [29, 30]. Systems demand reliable devices with specific operating points on demand, yet no systematic framework exists for achieving this at the hardware level.

For a single transneuron there can be up to 5 parameters affecting its dynamics, each interacting non-linearly through a set of coupled stochastic differential equations. The resulting parameter space is partitioned by bifurcation boundaries into qualitatively distinct dynamical regimes, meaning small parameter changes can produce discontinuous jumps in device behaviour. This renders traditional optimisation strategies such as gradient descent or grid search unreliable or intractable. At scale, this problem compounds combinatorially, and the curse of dimensionality makes optimisation impossible regardless of available hardware. Whilst significant progress has been made in optimising neuromorphic systems at the network level [31, 32, 33], device level parameter optimisation remains largely unaddressed. Existing approaches assume well characterised, controllable hardware components rather than providing the tools to achieve that characterisation. This work directly addresses that gap. First, an investigation of a deterministic reduction of the transneuron, establishing its suitability as a tractable analogue of the full stochastic system for the purposes of analysis and optimisation. Second, we introduce a machine-learning-assisted co-design framework utilising a surrogate multi-layer perception model to accelerate inference with proximal policy optimisation (PPO) to navigate the nonlinear parameter space. Together, these provide a scalable, sample-efficient template for the automated configuration of neuromorphic hardware, moving beyond manual device characterisation toward autonomous parameter optimisation, and bridging the gap between device physics and deployable neuromorphic systems.

Methods

Stochastic Transneuron Model

A visualisation of the transneuron device [26] is shown in Fig.1. Adopting the framework established by Midya et al. a phenomenological model composed from a set of coupled non-linear stochastic differential equations (Eq.1a-c) was derived. It considers the coupling between resistive heating and charge transport. These processes are dependent on resistance, which under certain assumptions, has been shown to be influenced by the stochastic diffusion of a few Ag particles occupying the insulated gap (L) between a nearly formed filament tip and a Pt anode [35]. The model relies on a

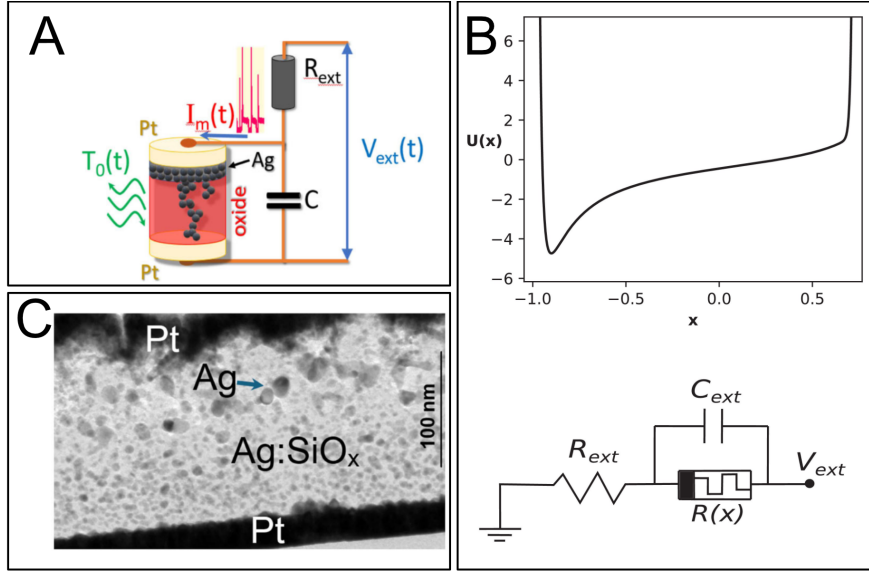


Figure 1: **Circuit diagram of memristor based transneuron device** **A** A diagram of the transneuron device incorporating a schematic of the underlying mechanism of a diffusive volatile memristor. **B** A plot of the phenomenologically derived chemical potential experienced by the a nonparticle moving in the within an insulated gap between the tip of a conducting filament and the Pt anode of the device. **C** Microscopic image of a $PtAgSiO_x$ diffusive memristor, the top Pt terminal shows the cathode whilst the bottom terminal shows the anode. **A** and **C** are adapted from [26]. **B** is reprinted with permission from [34].

core assumption that the relaxation time for mechanisms describing the electron dynamics ($\sim$ fs) are significantly faster than (i) thermal diffusion and (ii) the dynamics of the nanoparticles themselves ($\sim$ ps-ns). This separation by several orders of magnitude justifies a quasi-static approximation [36]. Furthermore, prior works have shown that a single particle consideration retains the qualitative behaviours of the device whilst simplifying numerical calculations [34, 37]. Under this simplification, the description of resistance $R(x)$ as a function of the instantaneous dimensionless particle position $x \in [-1, 1]$ is given by;

$$R(x) = R_t(e^{\frac{(L/2+x)}{\lambda}} + e^{\frac{(L/2-x)}{\lambda}}) = R(x) = R_0 \cosh\left(\frac{x}{\lambda}\right).$$

Here, R_t and R_0 represent the maximal and minimal resistance respectively. Maximal resistance R_t is observed when the particle remains close to the filament tip/terminal ($x = \pm 1$), and minimal resistance R_0 when the particle lies at the centre of the insulating gap ($x = 0$). Exponential terms with tunnelling factor λ describe the ease of charge transport through the insulated gap L via tunnelling effects as a result of the particles instantaneous position x .

$$\eta \frac{dx}{dt} = -\frac{\partial U(x)}{\partial x} + q \frac{V}{L} - \sqrt{\eta D \xi_i(t)}, \quad (1a)$$

$$\frac{dT}{dt} = \frac{V^2}{C_{th} R(x)} - \kappa(T - T_0), \quad (1b)$$

$$\tau \frac{dV}{dt} = V_{ext} - \left(1 + \frac{R_{ext}}{R(x)}\right)V. \quad (1c)$$

The evolution of x is modelled via a Langevin approximation where it is assumed that the particle is adiabatically damped, allowing inertial effects to be neglected and a reduction to a first order differential equation. Eq.(1a) is composed of forces determined by (i) $U(x)$ a phenomenologically motivated potential chosen to reflect the qualitative features of the energy landscape within the SiO_2 matrix (see Fig.1B) [37], (ii) a Coulomb force qV/L (q representing the effective charge of

a particle and V the electric potential) and, (iii) a stochastic Langevin force. The random thermal fluctuations acting on the particle are modelled using a Gaussian white noise correlation function $\xi_i(t)$ with zero mean and delta-correlation in time, implying a Markovian/memoryless process. The damping coefficient η characterises the extent of drag force experienced by the particle due to its suspension within the viscous SiO_2 matrix. The presence of η on the left hand side of Eq.(1a) can be attributed to the overdamped approximation from which this model is built upon. Also present in the Langevin force term is the diffusion coefficient D which is directly set by the thermal properties of the system. It is described by the Einstein-Smoluchowski relation for a particle confined to 1 dimension $D = 2k_bT$.

The rate of change of temperature T (Eq.(1b)) is expressed as the balance between thermal generation and thermal dissipation, modelled through Joule heating and Newton cooling terms respectively. The coefficient C_{th} represents the thermal capacitance, a material-specific term describing the energy required per unit change in temperature. Thermal dissipation arises from the consideration of an external bath, with temperature T_0 with which the device interacts. The rate of thermal dissipation is given by the product of temperature difference with κ , a cooling constant describing the characteristic rate of decay for a thermal relaxation process.

An equation for voltage [Eq.(1c)] is derived through the application of Kirchhoff's current law on the transneuron circuit shown in Fig.1B. The characteristic time constant $\tau = R_{ext}C_{ext}$ is analogous to that of an LC circuit where R_{ext} and C_{ext} represent the values of resistance and capacitance for the non-memristive components. An external voltage V_{ext} represents a global potential bias placed across the device. Within this work we consider cases when V_{ext} is simply a constant, and when V_{ext} is a harmonic oscillator with a DC offset [see Eq.(3)]. This is used to investigate the synchronous response of the device under the unidirectional influence of harmonic forcing. A notable omission from Midya et al's stochastic model is a term modelling the external voltage network noise [26]. It has not been incorporated here as this study is predominantly focused on the behaviour of a single transneuron device in isolation.

Deterministic Transneuron Model

Reducing this model to a deterministic set of differential equations is motivated by two primary factors. First, the removal of noise allows us to differentiate between noise-induced behaviours and self-sustained system dynamics. This distinction is vital for characterising regimes that offer tunable control over device behaviour. To support this, prior work using deterministic reductions of the compact memristor model [(Eq.(1a-c))] have shown spiking to be an intrinsic property of the system rather than a stochastic artefact [34, 37]. Second, the numerical integration of stochastic systems is significantly more costly than that of deterministic systems. This efficiency is particularly advantageous when working with Machine Learning and Reinforcement Learning techniques, which are known to struggle in noisy data environments [38] and benefit from the increased interpretability of a minimal baseline model. In the absence of network fluctuations, the primary source of noise is the stochastic motion of the Ag particle. This motion, predominantly driven by thermal effects, can be deterministically approximated via thermophoretic and thermoelectric forces [39, 40].

Under the assumption that the Ag particle represents the system's slowest dynamics, steady state solutions can be substituted into the original equations. As a result, an effective force acting on the particle arising from a spatially dependent thermal gradient T' can be derived. Through the coupling of temperature and voltage via the Joule heating term in Eq.(1b), an equation for the spatially dependent voltage gradient V' is required to close the system. Through this reduction, a set of 4 purely deterministic coupled ordinary differential equations are derived;

$$\eta \frac{dx}{dt} = -\frac{\partial U}{\partial x} + q \frac{V}{L} - q_T T', \quad (2a)$$

$$\frac{dT'}{dt} = \frac{2VV'R - V^2 R'}{C_{th} [R(x)]^2} - \kappa T', \quad (2b)$$

$$\tau \frac{dV}{dt} = V_{ext} - \left(1 + \frac{R_{ext}}{R(x)}\right) V, \quad (2c)$$

$$\tau \frac{dV'}{dt} = -\left(1 + \frac{R_{ext}}{R(x)}\right) V' + \frac{R_{ext} R'}{[R(x)]^2} V. \quad (2d)$$

The equation of motion for the particle now has its stochastic thermal force replaced by a deterministic thermophoretic force $q_T T'$ [41]. This is analogous to the electrostatic force exerted on a particle with charge q in a spatially dependent electromagnetic field only instead, 'charge' is described by a phenomenological 'thermal charge' (q_T) and the potential is determined by the temperature gradient T' . The spatial derivative of resistance $R'(x)$ appears in Eqs.(2b,d) and is given by taking the derivative of $R_0 \cosh(x/\lambda)$ w.r.t x , resulting in,

$$R' = \frac{R_0}{\lambda} \sinh\left(\frac{x}{\lambda}\right)$$

For all simulations in this work, a normalisation of Eqs.(2a-c) was used. A detailed qualitative justification for this reduction, as well as details regarding the dimensionless substitution, can be found in the Supplementary Information of the work by Midya et al. [26].

Numerical Methods

4th Order Runge-Kutta Integration

Due to the presence of non-linearity in Eqs.(2a-d), much of the in the work relies on the use of numerical integration. The classical fourth-order Runge-Kutta (RK4) method [42] was used to perform numerical continuation analysis alongside investigations into phase-locking synchronisation in the case of a time-dependent external voltage V_{ext} (see. Eq.[3]). RK4 is a fourth-order iterative, explicit, single-step method used to solve initial value problems of the form,

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0$$

Here, y is a vector and represents the state of our system $[x, T', V, V']$. The initial values of y_0 at time t_0 are chosen at the start of the simulation. A fixed time step size h is chosen and can be used to find an approximation for y at time $t + h$ using;

$$\begin{aligned} k_1 &= f(t_n, y_n), \\ k_2 &= f\left(t_n + \frac{h}{2}, y_n + k_1 \frac{h}{2}\right), \\ k_3 &= f\left(t_n + \frac{h}{2}, y_n + k_2 \frac{h}{2}\right), \\ k_4 &= f(t_n + h, y_n + h k_3), \\ y_{n+1} &= y_n + \frac{h}{6} (k_1 + 2k_2 + 2k_3 + k_4), \\ t_{n+1} &= t_n + h. \end{aligned}$$

This can be iteratively applied for any number of steps n . A step size of $h = 1 \times 10^{-3}$ was used for the majority of simulations. However, in regions with fast dynamics, the step size was decreased up to $h = 1 \times 10^{-4}$. In both cases, step size independence was verified by confirming that halving the step size produced negligible change in the computed trajectories. For simulations of time series data and parametric sweeping, standard initial conditions which reflected the unpowered state of the device, where the particle remains close to the filament tip, were used; $[x_0, T'_0, V_0, V'_0] = [-0.9999, 0, 0, 0]$. Simulations numerical integration were implemented in C++. Communication between python and C++ code as accomplished using pybind11 library.

Measuring Spikes

Spikes are discrete events which occur within neuronal systems. When presented with time series data obtained through numerical integration, the number of spikes in a given window of time can be measured using a one-way Poincaré map. To ensure robustness against numerical noise and prevent multiple counts for the same spike, the mapping was implemented using a logic-based state machine. A spike is registered when the trajectory of state variable x crosses a threshold x_{thresh} with a positive gradient ($\dot{x} > 0$). After a successful detection, the counter is locked until the state variable x falls below the threshold with a negative gradient ($\dot{x} < 0$). In addition, the time at which each spike occurred was also recorded.

Spike train statistics

A spike train can be defined as a discrete sequence of spikes produced consecutively within a given time interval. Spike train statistics provide metrics quantifying the correlations between consecutive spikes within spike trains. A foundational metric for spike train statistics is the inter-spike interval (ISI). This is a measure of the time difference between two consecutively occurring spikes. Taking the ISI for all spikes in a spike train produces a distribution of ISIs which can be analysed using standard statistical measurements [43]. An example of this would be the calculation of mean firing rate (measured in Hz), equivalent to the reciprocal of average ISI for a given distribution.

Another metric frequently used in neuroscience is the coefficient of variation CV_1 . It is defined as the ratio between the standard deviation and the mean. This is a dimensionless value bounded between 0 and 1 and can be used to infer the global temporal regularity of a spike train. A CV_1 value of 0 indicates periodic firing, implying no variation in the ISI distribution. In the opposite case, when $CV_1 = 1$, the train can be considered a Poisson process where spikes occur randomly and independently of one another. As the standard CV_1 measures the global variability of an entire time series, it is susceptible to slow changes in mean ISI. To solve this, a second measure for the coefficient of variation CV_2 is used. It measures local variations in ISI by looking only at neighbouring spikes. For a sequence of ISIs $[I_1, I_2, \dots, I_n]$ it can be approximated as the average of local CV_2 values,

$$CV_2 = \frac{1}{N-1} \sum_{i=1}^{N-1} \frac{2|I_{i+1} - I_i|}{I_{i+1} + I_i}.$$

Unlike CV_1 , CV_2 is bounded between 0 and 2 however, its interpretation is much the same. Values between 0 and 1 ($0 \leq CV_2 \leq 1$) can be understood in same manner as CV_1 . Values within the bounds $1 \leq CV_2 < 2$ indicate "bursty" behaviour, with wide variety between neighbouring ISIs.

These tools were used to measure the regularity of firing across a range of parameter configurations as well as during synchronisation. Certain regions demonstrated a behaviour akin to 'bursting' in biological cells [44]. In such cases, the the two timescales for these events were measured independently and are subsequently referred to as the 'inter-burst' and 'intra-burst' intervals. Inter-burst refers to the temporal correlation between 'bursting' events and intra-burst, the spikes which occur within these 'bursting' events. **Continuation Analysis**

Numerical continuation methods [45] were used to follow how the trajectories of the system [Eqs.(2a-d)] evolved as the parameters V_{ext} and R_{ext} changed. This was employed to create two-dimensional

regime maps in the (V_{ext}, R_{ext}) parameter plane. The systematic parameter sweep followed a row-major ordering, starting from the bottom left and completing an ascending and descending sweep of V_{ext} before incrementing R_{ext} and repeating this procedure. Forward and backward sweeps were necessary to investigate the hysteretic qualities of the system. It is worth clarifying that this method only presents the hysteretic behaviour of V_{ext} and does not account for hysteresis in R_{ext} . Initial conditions $([x_0, T'_0, V_0, V'_0] = [-0.9999, 0, 0, 0])$ were set at $V_{ext,0} = 1$ and $R_{ext,0} = 0$, after which, the final state at t_{max} is taken as the initial state for the next parameter in the sample set. The external voltage V_{ext} was sampled from a logarithmic series to maintain a constant sampling density across multiple orders of magnitude. An exception to this is found in Fig.3C. Measurements such as x_{max} , x_{min} , $\langle x \rangle$, and firing rate were used to distinguish between stationary, spiking, and bursting regimes. The total range of parameter values explored is summarised as:

- External Voltage (V_{ext}): Logarithmically sampled from 1.0V to 2000.0V with a typical density of 200 points.
- External Resistance (R_{ext}): Varied linearly from 0.0 Ω to 2000.0 Ω with a typical step size of $\Delta R_{ext} = 50.0 \tilde{\Omega}$.

The map in Fig.2 was produced by compiling data from several different parameter continuations, including upward and downward sweeps, using nearest neighbour interpolation implemented using `scipy.interpolate` [46].

Measuring Synchronisation

To evaluate the computational capacity of the deterministic transneuron model, we investigated its ability to act as a selective frequency filter through entrainment [47]. Synchronisation under unidirectional master-slave harmonic forcing was inspected using a time dependent external voltage expressed as,

$$V_{ext}(t) = V_{DC} + V_{AC} \cos(\omega t). \quad (3)$$

The external voltage is decomposed into a constant dc offset (V_{DC}) and a harmonic AC signal with amplitude V_{AC} and frequency ω . By subjecting the system to a periodic external stimulus, the Arnold Tongue can be mapped out. This is the region of parameter space where the neuron's firing rate is commensurate with that of the driving oscillator.

The following methodology details the extraction of higher-level metrics which provide quantitative benchmarks for the neuron's selectivity. The winding number, which measures the frequency difference between two coupled oscillators, was taken as the ratio between mean firing rate and driving frequency ω . The 1:1 phase locking region was identified by looking at the set of driving frequencies ω with a coinciding winding number of one, manifesting as a plateau. Several metrics about this region were calculated;

- Width: The range of frequencies for which 1:1 phase locking occurred. A smaller width implies a more selective (narrower) response.
- Centre of phase locking: The frequency at the centre of the phase locking region, representing the optimal tuned frequency of the neuron.
- Depth: The average difference between Coefficient of Variation (CV_2) for frequencies within the phase locking region versus those immediately adjacent. This quantifies the distinguishability in firing regularity of the neuron in the locked state from the unlocked state.

Measurements of the level of synchronisation were taken across parametric sweeps of ω . This differed from the continuation described previously in that the initial conditions for each parameter combination started from the same state vector $([x_0, T'_0, V_0, V'_0] = [-0.9999, 0, 0, 0])$ to ensure

independence between samples. For each frequency, a transient period, where measurements were not taken, was incorporated to allow the system to reach a stable limit cycle. To maintain numerical consistency, the transient and measurement window lengths were based on the occurrence of a fixed spike count of 200 and then 500 spikes for the transient and measurement period respectively. To mitigate the high computational cost of creating large sample sets and maps, the number of recorded spikes was decreased systematically whilst monitoring the stability of the width, centre, and depth metrics. This optimisation, combined with parallelisation, allowed for fast and efficient sampling of the neurons synchronous behaviour. It was found that the extraction of these higher level metrics (width, centre, and depth) effectively filtered out low-level numerical noise, ultimately allowing a 99% reduction in simulation time without sacrificing accuracy (see Appendix Table.A2).

Proximal Policy Optimisation (PPO)

Proximal Policy Optimisation is an on-policy gradient algorithm designed to optimise a clipped objective reward [48]. It uses an actor critic reinforcement learning (RL) architecture to generate a static policy vector field for any given environment. In this framework, an actor (the PPO algorithm) interacts with an environment in which observations can be taken and actions can be made. A surrogate critic model evaluates the actions taken by the actor by estimating the reward of the current state. Motivation behind the application of this approach towards the optimisation of artificial memristive neurons lies in it's ability to provide stable, deterministic, and adaptive policies for complex and non-linear environments. Envisioned is a framework with the ability to algorithmically bridge desired neuromorphic computational characteristics to optimal fabricable hardware parameters. An integral component for a successful RL implementation is the reward function, a mathematical formula that assigns a scalar value to actions taken by the agent. The reward function serves as the objective 'landscape' through which PPO learns to navigate, with the goal of converging towards the maxima of the function. In this study, the reward function was designed to optimise the width of the 1:1 phase locking region whilst providing control over the frequency for which the neuron is selective for. The final reward (R) is expressed as a negative sum of two primary penalties, a frequency error penalty and a gated width penalty. To ensure the system centres its selectivity near a specified target frequency f_{target} a pseudo Huber loss [49] was used to provide a reward with constant gradient far from the goal and differentiable reward close to the goal, $\epsilon_f = |f_{target} - f_{centre}|$:

$$P_f = k_f(\sqrt{\epsilon_f^2 + \delta^2} - \delta)$$

where k_f scales the penalty and δ acts as a smoothing parameter to ensure the gradient remains well-behaved near zero error. A critical feature of the reward function is a Gaussian [50] applied to the selectivity width (w). This is used as a gate [51] to ensure that the agent is only rewarded for narrowing the selectivity width once the system is already operating near the target frequency:

$$P_w = k_w w \exp(-\frac{\epsilon_f^2}{2\sigma_f^2})$$

The term σ_f defines the "radius of interest". If the frequency error is large, the width penalty is suppressed, allowing the agent to focus on achieving the correct desired selective frequency f_{target} before becoming trapped in local minima where selectivity is sharp but incorrectly positioned. The total reward returned to the agent is the negative sum of these components: $R = -(P_f + P_w)$. Whilst this specific optimisation problem only incorporated frequency error and width into the reward function for interpretability and simplicity, it is plausible to imagine the incorporation of more signals into the reward function.

One notable limitation encountered when trying to use PPO was it's sample inefficiency. When applied to optimise synchronisation across the parameter space of V_{ext} and R_{ext} , which as outlined in the sections above required parametric sweeps of driving frequency ω for each parameter

combination, the average frame rate (the time to make an observation, create an action and evaluate that action) exceeded 1 second. This made the use of PPO for this system computationally unfeasible and highly inefficient. This motivated the development of a surrogate model with the ability to reflect the general dynamics of the system at a vastly reduced wall-time inference cost. The implementation of PPO in the Stable-baselines3 python library developed by openAI was used [52]. To create the environment, the Gymnasium python framework was used [53]. For specific hyperparameters and details see Table.A5 in the Appendix.

Surrogate Model

The surrogate model was implemented using a simple MLP architecture with 2 inputs, V_{ext} and R_{ext} , 1 hidden layer with 64 neuron, and 5 ReLU activation outputs. A total of 512 points of training data was produced using Latin Hyper Cube sampling (LHS), a stochastic sampling approach which ensures, despite the fact that sampling is random, each parameter dimension is uniformly represented regardless of sample size. This method allows for efficient sampling of high dimensional datasets at low sampling rates [54]. To ensure the surrogate model remained physically consistent with the transneuron model, the output layer and data scaling was selected to prevent the generation of unphysical values (e.g. negative metrics). Min-Max scaling was used to bound all data within the normalised range [0,1]. This was combined with a Rectified Linear Unit (ReLU) activation function on the output layer to enforce non-negativity. This architectural constraint ensures the PPO agent is only exposed to physically plausible observations during training, improving the stability of the optimisation process. Validation of the surrogate model against the ODE integration model was carried out, providing the results shown in Tab.1. Inside 1:1 phase locking regions, CV tends to 0, skewing the mean absolute percentage error (MAPE). Therefore, we propose the normalised mean square error (N-MSE) as a better value for determining the validity of the surrogate model. The average N-MSE across all metrics was found to be 1×10^{-3} , providing sufficient grounds for the use of the surrogate model for training of the PPO model. As a result, use of the surrogate model vastly increased the speed of inference for the PPO algorithm, ultimately resulting in an average of ≈ 1500 fps during training. Architectural details of the specific mlp implementation and a heatmaps comparing the surrogate and ODE model can be found in Fig.A1 and Table.A4 in the Appendix.

Table 1: Surrogate Model Performance Metrics

Metric	N-MAE	N-MSE	MAPE (%)
Width	0.0260	0.0011	2.00%
CV1	0.0264	0.0029	938.79%
CV2	0.0148	0.0009	489.94%
Centre of Phase Locking	0.0045	0.0000	0.54%
Depth	0.0068	0.0002	3.94%

Results and Discussion

This section details the various results obtained through a culmination of the techniques and methods previously described. The findings obtained fit broadly under two categorical distinctions; a characterisation and mapping of the deterministic model, and the algorithmic optimisation for use as a selective computational unit.

To better understand the various dynamical regimes, a comprehensive mapping was constructed. Fig.2 is a collation of the domain knowledge obtained through continuation simulations and shows various plots demonstrating five distinct dynamical behaviours of the device across the (V_{ext}, R_{ext}) parameter plane. To begin with, the full regime map demonstrates a separation of roles between the two parameters. We can see that V_{ext} acts as the primary bifurcation parameter, driving the system through its sequence of dynamical transitions. As V_{ext} increases from low to high values the device progresses through a consistent and ordered sequence of regimes. Notably, limit cycles (indicated by hatched regions) are conditional upon a minimum resistance threshold of $R_{ext} > 220\Omega$). Above which the fundamental ordering of behaviours along the voltage axis is preserved for all plotted values.

The external resistance R_{ext} then acts as a modulating parameter that does not change the qualitative ordering of the regimes but controls how much of the V_{ext} axis each regime occupies. Interestingly, the spiking regimes I and III in Fig.2A exhibit a constant width when projected onto the logarithmic V_{ext} axis. The width ratio W_r is defined as the ratio between the upper and lower bifurcation thresholds bounding each spiking regime on the V_{ext} axis. Values for the width ratios of regimes I and III were calculated, resulting in $W_r^I = 6.5 \pm 0.2$ and $W_r^{III} = 2.2 \pm 0.2$. Furthermore, the rightward drift of each regime, whilst distorted by the logarithmic axis, was found to be linear (shown more clearly in Fig.3). Overall, this supports the use of R_{ext} as a tunable parameter with the ability to control the width of regimes and the voltage threshold at which spiking occurs (V_{th}). However, it must be taken into consideration that changing the voltage threshold V_{th} for spiking in regime I, through modulation of R_{ext} , requires subsequent scaling of the voltage input.

Another aspect of this system are it's hysteretic and bi-stable properties. This is clearly illustrated by Fig.2A through the asymmetry between the upper and lower map. The result validates and expands upon the findings of Midya et al. [26] by confirming the deterministic reductions ability to recreate the qualitative behaviours of the stochastic model and by extension, the physical transneuron device. Here R_{ext} also plays a role of switching between two forms of hysteresis. The first is labelled as regime IV and is better demonstrated by Fig.2C. This form of hysteresis exists for $R_{ext} < 900\Omega$ and is denoted specifically for the overlapping of a stable fixed point with the spiking region I. Above this resistance threshold the stable fixed point is annihilated and the hysteresis ceases, causing the upwards and downwards trajectories to coincide.

Also presented, is a previously undiscussed second tier of hysteresis. In Fig.2C it can be seen that upon the transition from regime III to V at $V_{ext} = 300V$ the particle enters a stable fixed point attractor from which it does not leave until voltage is reduced to $V_{ext} = 30$. The attractor sits at $x \approx 1$ and can be physically interpreted as a pinning of the particle to the Pt anode terminal through inter-facial electrostatic forces [55] and has been exploited previously to create synaptic emulators using similar diffusive memristors [56]. In this realisation of the phenomena, bursting in regime III is effectively silenced. Furthermore, the feature is analogous to certain biological neurons which exhibit the ability to suppress their output as an intrinsic feature of the neuron itself and not as a result of synaptic suppression [57, 58]. As a result of these two overlapping hysteresis loops, within the bounds $V_{ext} = [30, 95]$, three coexisting stable fixed point attractors exist. The presence of tristability is documented in biological neurons, often linked to neuron based information storage [59, 60] and has also been identified as a desirable feature for neuromorphic computing applications [61]. This biological similarity furthers motivations behind the transneuron device for use in unconventional forms of brain computation.

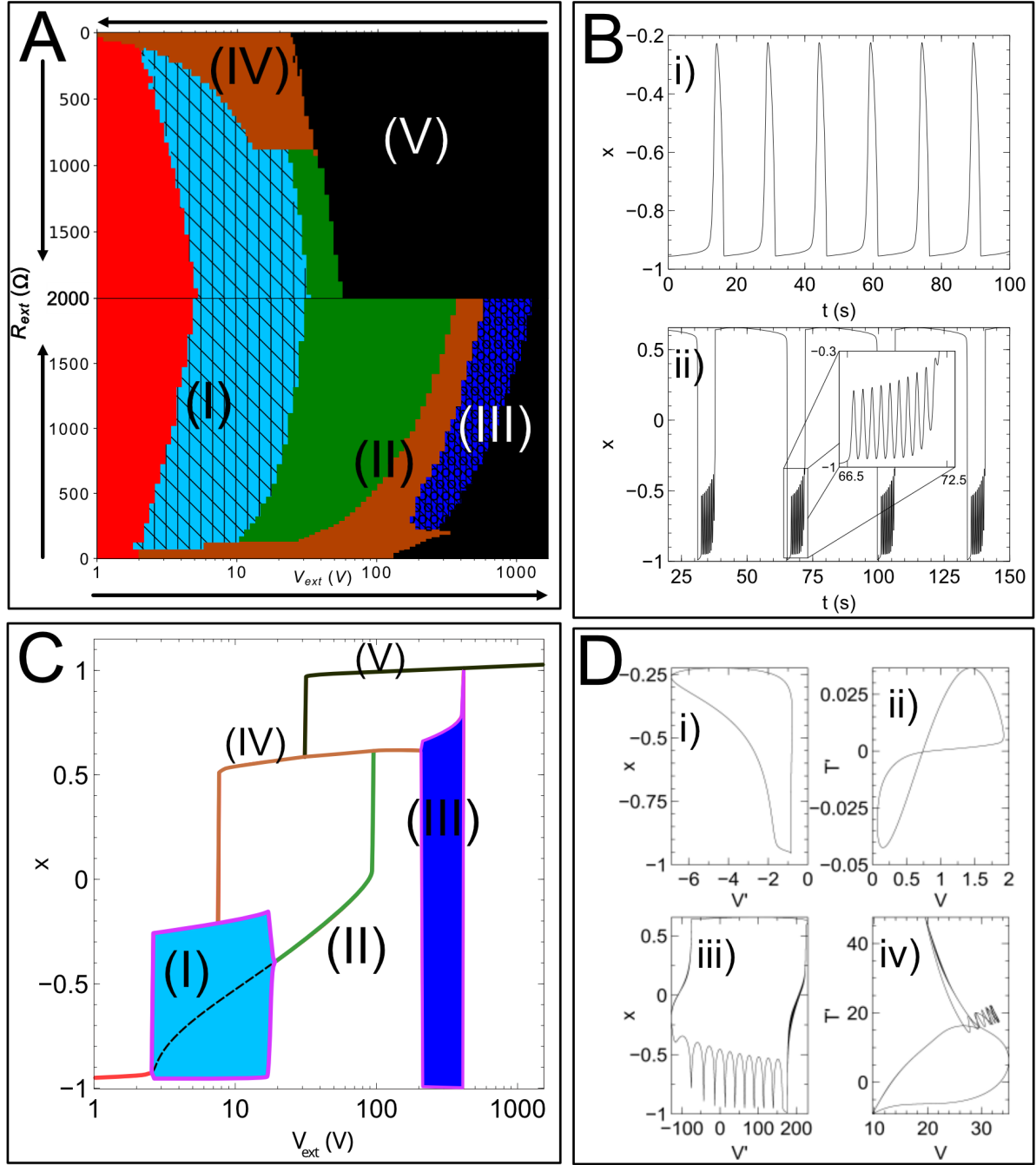


Figure 2: **The dynamical regimes of the deterministic model:** **A** A regime map on the (V_{ext}, R_{ext}) parameter plane demonstrating various distinct dynamical regimes. Hysteresis is highlighted by the asymmetry between the lower and upper halves, with black arrows indicating the direction of parameter continuation. Stable fixed point regimes are indicated by solid colours whilst hatching denotes the presence of a limit cycle. The various regimes have been numbered and correspond to: I A periodic spiking regime, II A region with two distinct stable fixed points, III A limit cycle regime demonstrating 'bursting' behaviour, IV A hysteretic stable fixed point which coexists with regime I, V A stable fixed point where the particle is pinned to the Pt anode of the device. **B** Time series plots showing the behaviour of state variable x in regime I (top) and III (bottom) obtained through numerical integration of Eqs.(2a-d). The inset in subplot ii) highlights intraburst oscillations. **C** A one dimensional slice (taken at $R_{ext} = 500$) of the full regime map (**A**), demonstrating the hysteric trajectory of x for an upward and downward continuation in the range $V_{ext} = [1, 1500]$. The colours and labels correspond with the same regimes as the full map. An unstable equilibrium within regime I is shown by a black dotted line. **D** Phase portraits showing the difference between limit cycles in regions I [subplots i)&ii)] and III [subplots iii)&iv)].

To provide more evidence of the deterministic reduction reproducing features of the stochastic model, Fig.2D shows the qualitative difference between regime I and regime III attractors. In regime I, the system exhibits a simple, isolated limit cycle, corresponding to regular periodic spiking in the time domain as a result of the charge and discharge cycle of the external capacitor (Fig.2B). Conversely, regime III is characterised by a complex multi-looped trajectory with two characteristic timescales. This behaviour is analogous to tonic bursting modes in biological neurons, one example are so called “chattering” superficial pyramidal neurons found within the visual cortex of monkeys and cats [58] and thalamic neurons [62]. It has previously been suggested that the fast timescale corresponds to the rapid spiking oscillations of the particle whilst the slow timescale is dominated by the heating and cooling cycles of the device [34]. This distinction has direct consequences for the frequency response of the device, which is explored in the following section.

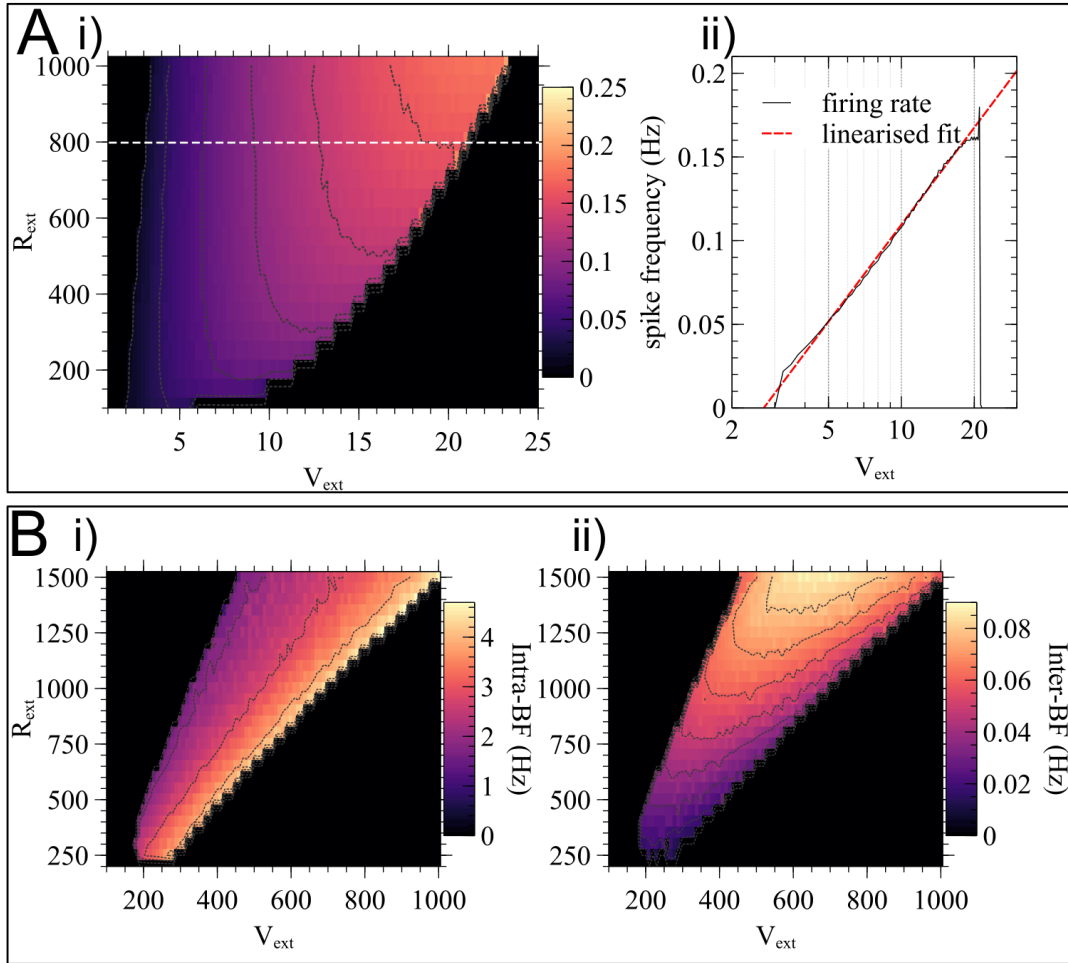


Figure 3: **Heatmaps of spiking rate as a function of the (V_{ext}, R_{ext}) parameter plane:** **Ai)** A linear colour heatmap showing the spiking frequency in Hz (also referred to as firing rate) within a portion of regime I against V_{ext} on the x axis and R_{ext} on the y axis. Regions where spiking does not occur naturally have a frequency of 0, indicated by the colour black. Grey dotted contour lines connecting points of equal value are included to better illustrate the landscape. **Aii)** A slice taken along the line $R_{ext} = 800$ (white dashed line on **Ai**) included for reference) showing the dependence of spiking frequency (referred to in the key as firing rate) on V_{ext} . The x axis is scaled logarithmically to illustrate the broadly logarithmic relationship between spiking rate and V_{ext} . The dashed red line represents a linearised line of best fit, calculated through a linear regression model. **Bi)** and **ii)** are linear colour heatmaps within a portion of regime III, demonstrating the frequency response for the two characteristic timescales; intra-burst frequency (frequency of oscillations within the burst), and the inter-burst frequency (frequency of the bursts) respectively. Regions of black indicate a lack of spiking behaviour.

Frequency response

To verify the tunability of the device, heatmaps quantifying the average firing rate for regimes I and

III were created. In Fig.3, it is shown that firing rate (spiking frequency) within the spiking regions can be precisely controlled using V_{ext} and R_{ext} . In regime I, an approximately logarithmic relationship was identified between spiking frequency and V_{ext} . Using linear regression on the linearised V_{ext} axis a general relationship for the spiking frequency $f_s(V_{ext}) = m \ln V_{ext} + c$ was obtained. Setting $f_s = 0$ and solving for the x-intercept yields an expression for the threshold voltage $V_{th} = e^{-c/m}$, below which spiking does not occur. Substituting back gives a more physically transparent form,

$$f_s(V_{ext}) = m \cdot \ln \left(\frac{V_{ext}}{V_{th}} \right),$$

where $m = 0.0837 \pm 0.016$ and V_{th} is determined by R_{ext} (fit parameters in Appendix Tab.A3). This can be further refined using a piece-wise approximation,

$$f_s(V_{ext}) = \begin{cases} 0 & V_{ext} < V_{th} \\ m \ln \left(\frac{V_{ext}}{V_{th}} \right) & V_{th} \leq V_{ext} \leq V_{end} \\ 0 & V_{ext} > V_{end} \end{cases}$$

where $V_{end} = W_r^I V_{th}$. Using the approximate value for the width ratio in regime I ($W_r^I = 6.5 \pm 0.2$), we obtain a compact empirical model for the spiking frequency at any given V_{ext} , provided we know the voltage threshold V_{th} . It should be noted that this expression breaks down quantitatively close to the boundary bifurcation points of regime I.

Furthermore, we provide evidence for the independent control of the two characteristic timescales responsible for bursting in regime III through tuning of V_{ext} and R_{ext} . It was found that V_{ext} is precisely responsible for the intra-burst frequency, whilst R_{ext} is predominantly responsible for the inter-burst frequency. The two observations present opportunity for a multiplexing scheme, in which distinct information streams are encoded simultaneously across different timescales. This encoding strategy has been documented in thalamic neural circuits [63] and has been theorised to optimise information transmission in bursting neural codes [64, 65]. The independent tunability of the two timescales provides a physical mechanism for such a scheme, suggesting regime III's capability for rich computational modes.

Optimisation of computation by selectivity

Having established the regime structure and frequency response of the deterministic transneuron model, we now investigate the usefulness of this model for exploring and optimising the capability to perform computation by selectivity. Selective responses to frequency stimulus in neurons has been documented as a mode of brain computation [66, 65] and was demonstrated by Midya et al. in the transneuron and stochastic transneuron model [26]. Here, we establish that the deterministic reduction preserves this capability. Presented in more detail within the methods, computation by selectivity in the deterministic model was investigated parametrically across a range of external forcing frequencies using a harmonic external voltage of the form $V_{ext} = V_{DC} + V_{AC} \cos(\omega t)$ [Eq.(3)].

It was found that the deterministic model clearly exhibits phase locking synchronisation [47], the effects of which can be seen in Fig.4. Plateaus in the winding number at 1 and 0.5 represent slices of the Arnold tongue, regions of 1:1 and 1:2 phase locking synchronisation. Around these regions we see CV_1 assume a 'bucket' shaped curve with a flat bottom at $y=0$ (Figs.4i)&ii)). We observe that CV_1 exhibits much steeper walls than CV_2 , suggesting that CV_2 may provide a smoother optimisation landscape. Therefore, the selective response of the neuron is encoded in the temporal regularity of it's spike train. The metrics discussed in the methods for synchronisation refer to the depth and width of this 'bucket' and serve as good optimisation targets for computing by selectivity. Having confirmed that the deterministic model supports selective frequency filtering, the question becomes how to efficiently navigate the parameter space to optimise these selectivity metrics for a given target, which motivates the PPO framework introduced next.

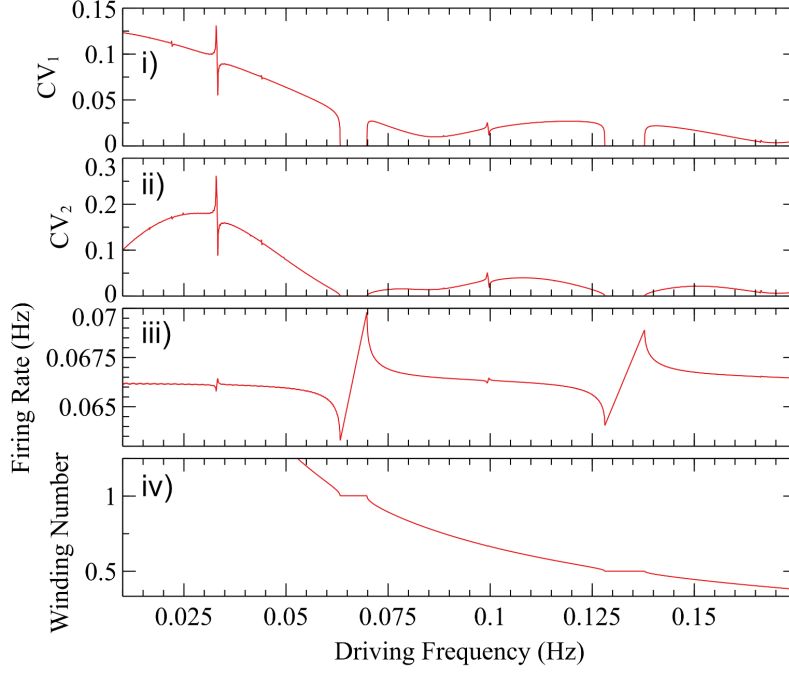


Figure 4: **Synchronous response of transneuron deterministic model:** A collection of plots showing measurements of synchronisation (y axis) for the deterministic transneuron model ($V_{ext} = 6, R_{ext} = 500$) under the forcing of a harmonic external voltage signal of the form $V_{ext}(t) = V_{DC} + V_{AC} \cos(\omega t)$, where $V_{DC} = 6, V_{AC} = 1$ and ω varied parametrically from 0.01-0.2 (Hz) in steps of $f_{step} = 1 \times 10^{-4}$. Measurements were taken over the course of 500 spikes after a transient period of 200 spikes. **i)&ii)** show the CV_1 and CV_2 , measurements of the temporal regularity of spiking. CV values of 0 indicate periodic firing as a result of phase locking frequency synchronisation. **iii)** compares the firing rate of the transneuron with the frequency of the forcing signal. **iv)** Shows a plot of winding number, the frequency ratio between to coupled oscillators, against forcing frequency ω . Plateaus at $y = 1$ and $y = 0.5$ indicate regions of 1:1 and 1:2 phase locking synchronisation respectively.

Table 2: Results of PPO Agent Computation by Selectivity Optimisation

Fixed Evaluation Parameters:					
$N = 225$, Target $f_{centre} = 0.06$ Hz, Max Steps = 100					
Optimisation Results					
Metric	Initial ($\mu \pm \sigma$)	Final ($\mu \pm \sigma$)	Delta	CV (%)	Improvement
V_{DC} (V)	—	5.5079 ± 0.0150	-0.2407	0.27	-4.19%
R_{ext} (Ω)	—	660.84 ± 6.80	+85.65	1.03	+14.89%
$ f_{err} $	0.0146 ± 0.0092	0.0002 ± 0.0002	-0.0144	—	+95.60%
Width (Hz)	0.0100 ± 0.0015	0.0087 ± 0.0002	-0.0013	—	-10.70%
Depth	0.0511 ± 0.0307	0.0380 ± 0.0004	-0.0131	—	-9.20%
CV1	0.0384 ± 0.0234	0.0000 ± 0.0000	-0.0384	—	-99.96%
CV2	0.0445 ± 0.0382	0.0000 ± 0.0000	-0.0445	—	-99.97%
Success Rate	100.0%	Conv. Speed	42.03 ± 14.8 steps	Jitter: 0.0045	

PPO Framework

Here we evidence a proof of concept framework for the algorithmic optimisation of the transneuron utilising ML and RL techniques. Using the domain knowledge obtained through the characterisation of the deterministic model, suitable set of parameters which provide a stable and interpretable base

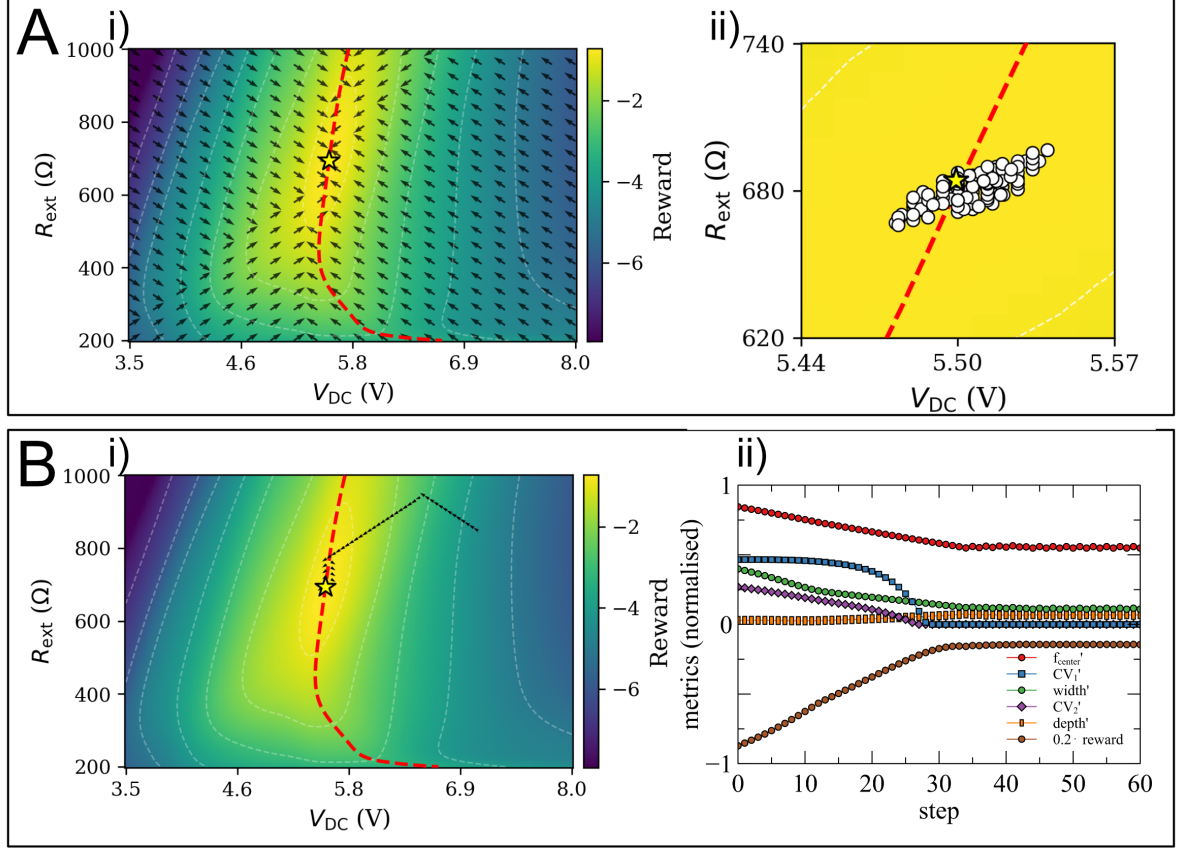


Figure 5: **PPO optimisation policy:** **A** Visualisation of the policy vector field upon the reward landscape and it's convergence upon the maximum reward on the (R_{ext}, R_{ext}) parameter plane. **Ai)** Arrows indicate the direction and scaled magnitude of the action the PPO agent would take for a given external voltage and resistance. In the background is a heatmap of the pseudo-Huber Gaussian gated reward landscape as seen by the PPO agent (see to methods section for more detail). A gold star indicates the maximum point of the reward landscape, serving as the goal of the PPO agent. The red dashed line running vertically central marks a target frequency isoline ($f_{target} = 0.06Hz$), connecting all points of equal spiking frequency. **Aii)** A zoomed plot of the reward landscape centred around the maximum reward. White dots represent the final actions of 225 agents initialised on an equally spaced 15×15 grid spanning the full range of the policy vector field shown in **Ai)**. **B** Contains plots showing the trajectory of a single agent initialised at $V_{ext} = 7.1V$ and $R_{ext} = 840\Omega$, **Bii)** shows the evolution of various normalised selectivity metrics ($[0,1]$) as a function of the agents steps. In order from top to bottom on the y axis at initialisation we have, f'_{center} , CV'_1 , $width'$, CV'_2 , $depth'$, $reward$, apostrophe's (') are used to indicate normalisation. In addition, the reward curve has been scaled by a factor of 0.2 to aid with visualisation.

case was devised. For a fixed target frequency ($f_{target} = 0.06$) and a range of device parameters $V_{ext} = [3.5, 8]$ and $R_{ext} = [150, 1000]$, five selectivity metrics were calculated across parametric sweeps of $\omega = [0.02, 0.1]$. It is known within the reinforcement learning community that implementations of plain PPO are often sample inefficient as a result of it's on-policy approach [67, 68]. To bypass this issue, as detailed in the methods, a surrogate MLP model was trained on 512 pairs of V_{ext} R_{ext} combinations and their five corresponding selectivity metrics. The use of the surrogate model improved the inference time significantly, making training the PPO model computationally feasible.

The policy generated after 500 iterations demonstrated a global understanding of the reward landscape, with agents able to convergence close to the maximum reward regardless of their initialisation. The policy vector field is shown in Fig.5Ai). The trajectory of a single agent is also included in Fig.5B to give an idea of the efficiency of convergence. The average speed of convergence across the whole reward landscape was found to be 42.03 ± 14.8 steps, indicating that the agent did not

learn optimal trajectories (as is clearly shown in Fig.5Bi)). It is understandable why this occurred seeing as distance was not penalised in the reward function. Despite this, it can be seen in Fig.5 that the optimisation of the selectivity metrics over the course of the agents trajectory was smooth, gradual, and quite linear. A comparison between the initial and the final metrics was performed for a set of 255 equally spaced agents initialised on a 15x15 grid. We found that the agent was highly capable of navigating the landscape, producing parameter combinations which converged on the target frequency with a mean absolute frequency error after convergence of $2 \times 10^{-4} \pm 2 \times 10^{-4}$. This analysis was conducted on the original ODE simulation using the voltage and resistance values provided by the PPO agent to evidence the validity of the surrogate model. A table collecting statistical measurements of the policies convergence has been constructed (see Tab.2). Looking at this table more closely, we see that the width of the phase locking region was indeed minimised by a mean percentage of 10.7%. It should be noted that the depth decreased during convergence, which has the effect of reducing selectivity. Seeing as it was not included in the reward function, this is not an unexpected result.

Meta analysis of the PPO agent showed a 100% success rate (taken as a the accomplishment of $|f_{err}| < 1\%$). The collective mean μ and standard deviation σ for the final $V_{ext} = 5.5079 \pm 0.0150V$ and $R_{ext} = 660.84 \pm 6.8\Omega$, these indicate a tight spread of agents around a globally set point. As discussed previously, the primary control parameter for spiking frequency in regime I is V_{ext} , this is reflected in the comparatively lower $CV = 0.27\%$. This suggests that the policy has learnt this global feature of the device, prioritising higher precision for V_{ext} over R_{ext} .

With regards to the quality of learning, the final explained variance of 0.997 (the critics ability to predict the returns of the environment) and extremely low Approx. KL divergence of 2.745×10^{-5} (a measure of how much the policy changes per update) indicate that the policy converged to a stable solution without catastrophic forgetting or unstable updates.

While this implementation remains a proof of concept, only optimising for a single target frequency within a specifically chosen range of V_{ext} and R_{ext} parameter combinations, the framework is designed to be generalised towards multi-objective optimisation incorporating additional selectivity metrics, model parameters and ultimately the stochastic model. The success of this model provides sufficient grounds to pursue further investigation into the development of this approach.

Conclusion

The utilisation of domain knowledge obtained through regime mapping and model characterisation was successfully used to inform the development of an ML and RL framework for the automated optimisation of a memristive transneuron for use as a selective frequency filter. This directly addresses the layer that is currently missing between device level characterisation and systems level deployment within the neuromorphic cohort.

It was found that a deterministic reduction of the transneuron introduced by Midya et al. [26] acts as a tractable analogue of the full stochastic model, preserving the qualitative dynamical behaviour necessary for meaningful analysis and optimisation. This enabled the application of numerical continuation and dynamical systems theory, exposing the tunable skeleton of the transneuron across the (V_{ext}, R_{ext}) parameter plane. The resulting regime characterisation provided evidence for the device's applicability to computation by selectivity and the multiplex encoding of information through two independently controllable characteristic timescales of bursting. A previously undiscussed tristable region, arising from two overlapping hysteresis loops was identified, enabling the dynamical silencing of bursting in regime III and providing the operational parameter conditions under which this occurs. Furthermore, an empirical logarithmic model for the spiking frequency as a function of voltage threshold was derived, offering a compact analytical expression for the frequency response of the device within regime I. A proof of concept PPO optimisation policy was trained using a surrogate MLP environment and evaluated against the original ODE simulation. For 225 agents initialised

across the full normalised parameter space, the policy converged reliably to the target frequency configuration of $f_{target} = 0.06Hz$, achieving an explained variance of 0.997 and an approximate KL divergence of 2.745×10^{-5} , indicating stable convergence.

Despite these findings, the limitations of what has been demonstrated must be considered. The deterministic reduction sacrifices stochasticity, obscuring a component of the transneurons real electrical behaviour that is increasingly understood to be functionally significant. Stochasticity has emerged as a fundamental and potentially intrinsic computational property of biological systems, and it is believed that neuromorphic hardware will benefit from reflecting this [69, 70, 71]. The PPO framework presented here optimises only two objectives within a carefully selected and controlled parameter window and it is not yet established how well the surrogate model generalises across the full parameter range of the device. These limitations define the immediate priorities for future development.

The broader value of this work lies in what it sets out to establish. Envisioned is a generalisable framework for multi objective optimisation and hardware software co-design, operating directly on the stochastic model across a much larger parameter space. Such a framework would provide the layer between device physics and functional neuromorphic systems that the field currently lacks [27]. Extending this to networked devices, initial steps would involve the optimisation of small functional neuron blocks with defined computational characteristics that can be systematically combined into larger neuromorphic networks. This work represents a first step along that trajectory, demonstrating that principled, interpretable optimisation of individual memristive devices is feasible, and has that potential to contribute towards designing scalable neuromorphic systems in the future.

Acknowledgements

I would like to acknowledge support of my friends and family. In addition, I would also like to acknowledge the nurturing guidance of my supervisors throughout the course of this project.

References

- [1] Gordon Moore. "Cramming More Components onto Integrated Circuits, Reprinted from Electronics, Volume 38, Number 8, April 19, 1965, Pp.114 Ff". In: Solid-State Circuits Newsletter, IEEE 11 (Oct. 1, 2006), pp. 33–35. DOI: 10.1109/N-SSC.2006.4785860.
- [2] Gordon Moore. "Progress In Digital Integrated Electronics [Technical Literature, Copyright 1975 IEEE. Reprinted, with Permission. Technical Digest. International Electron Devices Meeting, IEEE, 1975, Pp. 11-13.]" In: Solid-State Circuits Newsletter, IEEE 20 (Oct. 1, 2006), pp. 36–37. DOI: 10.1109/N-SSC.2006.4804410.
- [3] Laszlo B Kish. "End of Moore's Law: Thermal (Noise) Death of Integration in Micro and Nano Electronics". In: Physics Letters A 305.3 (Dec. 2, 2002), pp. 144–149. ISSN: 0375-9601. DOI: 10.1016/S0375-9601(02)01365-8.
- [4] Kenneth Flamm. "Measuring Moore's Law: Evidence from Price, Cost, and Quality Indexes". In: Measuring and Accounting for Innovation in the Twenty-First Century. University of Chicago Press, May 2019, pp. 403–470.
- [5] David Burg and Jesse H. Ausubel. "Moore's Law Revisited through Intel Chip Density". In: PLOS ONE 16.8 (Aug. 18, 2021), e0256245. ISSN: 1932-6203. DOI: 10.1371/journal.pone.0256245.
- [6] Martijn Koot and Fons Wijnhoven. "Usage Impact on Data Center Electricity Needs: A System Dynamic Forecasting Model". In: Applied Energy 291 (June 1, 2021), p. 116798. ISSN: 0306-2619. DOI: 10.1016/j.apenergy.2021.116798.

- [7] Sadasivan Shankar and Albert Reuther. "Trends in Energy Estimates for Computing in AI/Machine Learning Accelerators, Supercomputers, and Compute-Intensive Applications". In: 2022 IEEE High Performance Extreme Computing Conference (HPEC). Sept. 19, 2022, pp. 1–8. DOI: 10.1109/HPEC55821.2022.9926296. arXiv: 2210.17331 [cs].
- [8] Thomas N. Theis and H.-S. Philip Wong. "The End of Moore's Law: A New Beginning for Information Technology". In: Computing in Science & Engineering 19.2 (Mar. 2017), pp. 41–50. ISSN: 1558-366X. DOI: 10.1109/MCSE.2017.29.
- [9] John Shalf. "The Future of Computing beyond Moore's Law". In: Philos Trans A Math Phys Eng Sci 378.2166 (Jan. 20, 2020), p. 20190061. ISSN: 1364-503X. DOI: 10.1098/rsta.2019.0061.
- [10] Giacomo Indiveri and Shih-Chii Liu. "Memory and Information Processing in Neuromorphic Systems". In: Proceedings of the IEEE 103.8 (Aug. 2015), pp. 1379–1397. ISSN: 1558-2256. DOI: 10.1109/JPRDC.2015.2444094.
- [11] Xingqi Zou et al. "Breaking the von Neumann Bottleneck: Architecture-Level Processing-in-Memory Technology". In: Sci. China Inf. Sci. 64.6 (Apr. 27, 2021), p. 160404. ISSN: 1869-1919. DOI: 10.1007/s11432-020-3227-1.
- [12] Lianchun Yu and Yuguo Yu. "Energy-Efficient Neural Information Processing in Individual Neurons and Neuronal Networks". In: Journal of Neuroscience Research 95.11 (2017), pp. 2253–2266. ISSN: 1097-4547. DOI: 10.1002/jnr.24131.
- [13] Guillaume A. Rousselet, Simon J. Thorpe, and Michèle Fabre-Thorpe. "How Parallel Is Visual Processing in the Ventral Pathway?" In: Trends Cogn Sci 8.8 (Aug. 2004), pp. 363–370. ISSN: 1364-6613. DOI: 10.1016/j.tics.2004.06.003. PMID: 15335463.
- [14] Yuhai Tu and Wouter-Jan Rappel. "Adaptation in Living Systems". In: Annual Review of Condensed Matter 9 (Volume 9, 2018 Mar. 10, 2018), pp. 183–205. ISSN: 1947-5454, 1947-5462. DOI: 10.1146/annurev-conmatphys-033117-054046.
- [15] Danielle S. Bassett et al. "Dynamic Reconfiguration of Human Brain Networks during Learning". In: Proceedings of the National Academy of Sciences 108.18 (May 3, 2011), pp. 7641–7646. DOI: 10.1073/pnas.1018985108.
- [16] Pavan Kumar Enuganti et al. "Neuromorphic Computing and Applications: A Topical Review". In: WIREs Data Mining and Knowledge Discovery 15.2 (2025), e70014. ISSN: 1942-4795. DOI: 10.1002/widm.70014.
- [17] Nicholas J. Pritchard et al. A Bibliometric Review of Neuromorphic Computing and Spiking Neural Networks. Apr. 14, 2023. DOI: 10.48550/arXiv.2304.06897. arXiv: 2304.06897 [cs]. Pre-published.
- [18] H B Barlow. "Single Units and Sensation: A Neuron Doctrine for Perceptual Psychology?" In: Perception 1.4 (Dec. 1, 1972), pp. 371–394. ISSN: 0301-0066. DOI: 10.1068/p010371.
- [19] Leon O. Chua. "Memristor—The Missing Circuit Element". In: IEEE Trans. Circuit Theory 18.5 (1971), pp. 507–519. ISSN: 0018-9324. DOI: 10.1109/TCT.1971.1083337.
- [20] Dmitri B. Strukov et al. "The Missing Memristor Found". In: Nature 453.7191 (May 1, 2008), pp. 80–83. ISSN: 0028-0836, 1476-4687. DOI: 10.1038/nature06932.
- [21] Ke Yang et al. "Nonlinearity in Memristors for Neuromorphic Dynamic Systems". In: Small Science 2.1 (2022), p. 2100049. ISSN: 2688-4046. DOI: 10.1002/smssc.202100049.
- [22] Rui Yang, He-Ming Huang, and Xin Guo. "Memristive Synapses and Neurons for Bioinspired Computing". In: Advanced Electronic Materials 5.9 (2019), p. 1900287. ISSN: 2199-160X. DOI: 10.1002/aelm.201900287.
- [23] Wen Huang et al. "Memristive Artificial Synapses for Neuromorphic Computing". In: Nano-Micro Lett. 13.1 (Mar. 6, 2021), p. 85. ISSN: 2150-5551. DOI: 10.1007/s40820-021-00618-2.

- [24] Haider Abbas et al. "Emulation of Short-Term and Long-Term Synaptic Plasticity with High Uniformity in Chalcogenide-Based Diffusive Memristor Device for Neuromorphic Applications". In: Journal of Materials Science & Technology 216 (May 2025), pp. 99–107. ISSN: 10050302. DOI: 10.1016/j.jmst.2024.07.033.
- [25] Silvia Tolu. "2022 Roadmap on Neuromorphic Computing and Engineering". In: Neuromorphic Computing and Engineering (Jan. 1, 2022). DOI: 10.1088/2634-4386/AC4A83.
- [26] Rivu Midya et al. "Artificial Transneurons Emulate Neuronal Activity in Different Areas of Brain Cortex". In: Nat Commun 16.1 (Aug. 7, 2025), p. 7289. ISSN: 2041-1723. DOI: 10.1038/s41467-025-62151-9.
- [27] Dhireesha Kudithipudi et al. "Neuromorphic Computing at Scale". In: Nature 637.8047 (Jan. 23, 2025), pp. 801–812. ISSN: 0028-0836, 1476-4687. DOI: 10.1038/s41586-024-08253-8.
- [28] Shimeng Yu. "Neuro-Inspired Computing with Emerging Nonvolatile Memories". In: Proceedings of the IEEE 106.2 (Feb. 2018), pp. 260–285. ISSN: 1558-2256. DOI: 10.1109/JPROC.2018.2790840.
- [29] Jeffrey S. Vetter et al. "Abisko: Deep Codesign of an Architecture for Spiking Neural Networks Using Novel Neuromorphic Materials". In: The International Journal of High Performance Computing Applications 37.3–4 (July 1, 2023), pp. 351–379. ISSN: 1094-3420. DOI: 10.1177/10943420231178537.
- [30] Yingheng Tang et al. "Device-System End-to-End Design of Photonic Neuromorphic Processor Using Reinforcement Learning". In: Laser & Photonics Reviews 17.2 (2023), p. 2200381. ISSN: 1863-8899. DOI: 10.1002/lpor.202200381.
- [31] Nitin Rathie et al. "Exploring Neuromorphic Computing Based on Spiking Neural Networks: Algorithms to Hardware". In: ACM Comput. Surv. 55.12 (Mar. 2, 2023), 243:1–243:49. ISSN: 0360-0300. DOI: 10.1145/3571155.
- [32] Mishal Fatima Minhas et al. "Continual Learning With Neuromorphic Computing: Foundations, Methods, and Emerging Applications". In: IEEE Access 13 (2025), pp. 124824–124873. ISSN: 2169-3536. DOI: 10.1109/ACCESS.2025.3588665.
- [33] Seham Al Abdul Wahid, Arghavan Asad, and Farah Mohammadi. "A Survey on Neuromorphic Architectures for Running Artificial Intelligence Algorithms". In: Electronics 13.15 (Jan. 2024), p. 2963. ISSN: 2079-9292. DOI: 10.3390/electronics13152963.
- [34] Yury Ushakov, Alexander Balanov, and Sergey Savel'ev. "Role of Noise in Spiking Dynamics of Diffusive Memristor Driven by Heating-Cooling Cycles". In: Chaos, Solitons & Fractals 145 (Apr. 2021), p. 110803. ISSN: 09600779. DOI: 10.1016/j.chaos.2021.110803.
- [35] D.P. Pattnaik et al. "Temperature Control of Diffusive Memristor Hysteresis and Artificial Neuron Spiking". In: Phys. Rev. Appl. 19.2 (Feb. 24, 2023), p. 024065. DOI: 10.1103/PhysRevApplied.19.024065.
- [36] Sergey E. Savel'ev, Fabio Marchesoni, and Alexander M. Bratkovsky. "Mesoscopic Resistive Switch: Non-Volatility, Hysteresis and Negative Differential Resistance". In: Eur. Phys. J. B 86.12 (Dec. 2013), p. 501. ISSN: 1434-6028, 1434-6036. DOI: 10.1140/epjb/e2013-40966-4.
- [37] A. Akther et al. "Deterministic Modeling of the Diffusive Memristor". In: Chaos: An Interdisciplinary Journal 31.7 (July 1, 2021), p. 073111. ISSN: 1054-1500, 1089-7682. DOI: 10.1063/5.0056239.
- [38] Jingkan Wang, Yang Liu, and Bo Li. "Reinforcement Learning with Perturbed Rewards". In: Proceedings of the AAAI Conference on Artificial Intelligence 34.04 (Apr. 3, 2020), pp. 6202–6209. ISSN: 2374-3468. DOI: 10.1609/aaai.v34i04.6086.

- [39] Ming Wang et al. "Thermoelectric Seebeck Effect in Oxide-Based Resistive Switching Memory". In: Nat Commun 5.1 (Aug. 20, 2014), p. 4598. ISSN: 2041-1723. DOI: 10.1038/ncomms5598.
- [40] Suhas Kumar et al. "Conduction Channel Formation and Dissolution Due to Oxygen Thermophoresis/Diffusion in Hafnium Oxide Memristors". In: ACS Nano 10.12 (Dec. 27, 2016), pp. 11205–11210. ISSN: 1936-0851, 1936-086X. DOI: 10.1021/acsnano.6b06275.
- [41] Daniel B. Mayer et al. "Thermophoresis beyond Local Thermodynamic Equilibrium". In: Phys. Rev. Lett. 130.16 (Apr. 19, 2023), p. 168202. DOI: 10.1103/PhysRevLett.130.168202.
- [42] "Runge-Kutta Method". In: William H. Press et al. Numerical Recipes in C++: The Art of Scientific Computing. Cambridge, UK ; New York: Cambridge University Press, 2002, pp. 907–910. ISBN: 978-0-521-75033-2.
- [43] "Stochastic Activity in Integrate-and-Fire Models". In: Christof Koch. Biophysics of Computation: Information Processing in Single Neurons. USA: Oxford University Press, Inc., Sept. 2004, pp. 355–366. ISBN: 978-0-19-518199-9.
- [44] "Bursting Cells". In: Christof Koch. Biophysics of Computation: Information Processing in Single Neurons. USA: Oxford University Press, Inc., Sept. 2004, pp. 374–381. ISBN: 978-0-19-518199-9.
- [45] Emilio Freire and Alejandro J. Rodríguez-Luis. "Numerical Bifurcation Analysis of Electronic Circuits". In: Numerical Continuation Methods for Dynamical Systems: Path Following and Boundary Value Problems. Ed. by Bernd Krauskopf, Hinke M. Osinga, and Jorge Galán-Vioque. Dordrecht: Springer Netherlands, 2007, pp. 221–251. ISBN: 978-1-4020-6356-5. DOI: 10.1007/978-1-4020-6356-5_7.
- [46] Pauli Virtanen et al. "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python". In: Nat Methods 17.3 (Mar. 2020), pp. 261–272. ISSN: 1548-7105. DOI: 10.1038/s41592-019-0686-2.
- [47] Alexander Balanov et al. "N : M Synchronization of Periodic Oscillations". In: Synchronization: From Simple to Complex. Ed. by Alexander Balanov et al. Berlin, Heidelberg: Springer, 2009, pp. 121–148. ISBN: 978-3-540-72128-4. DOI: 10.1007/978-3-540-72128-4_6.
- [48] John Schulman et al. Proximal Policy Optimization Algorithms. Aug. 28, 2017. DOI: 10.48550/arXiv.1707.06347. arXiv: 1707.06347 [cs]. Pre-published.
- [49] Shaili Mishra and Anuja Arora. "A Huber Reward Function-Driven Deep Reinforcement Learning Solution for Cart-Pole Balancing Problem". In: Neural Comput & Applic 35.23 (Aug. 1, 2023), pp. 16705–16722. ISSN: 1433-3058. DOI: 10.1007/s00521-022-07606-6.
- [50] Jinyeob Kim et al. "Transformable Gaussian Reward Function for Socially Aware Navigation Using Deep Reinforcement Learning". In: Sensors 24.14 (Jan. 2024), p. 4540. ISSN: 1424-8220. DOI: 10.3390/s24144540.
- [51] Zetian Sun et al. Stabilizing Long-term Multi-turn Reinforcement Learning with Gated Rewards. Aug. 14, 2025. DOI: 10.48550/arXiv.2508.10548. arXiv: 2508.10548 [cs]. Pre-published.
- [52] Antonin Raffin et al. "Stable-Baselines3: Reliable Reinforcement Learning Implementations". In: Journal of Machine Learning Research 22.268 (2021), pp. 1–8.
- [53] Mark Towers et al. Gymnasium: A Standard Interface for Reinforcement Learning Environments. arXiv.org. July 24, 2024.
- [54] R. L. Iman. "Latin Hypercube Sampling". In: Encyclopedia of Statistical Sciences. John Wiley & Sons, Ltd, 2006. ISBN: 978-0-471-66719-3. DOI: 10.1002/0471667196.ess1084.pub2.
- [55] Solomon Amsalu Chekol et al. "Effect of the Threshold Kinetics on the Filament Relaxation Behavior of Ag-Based Diffusive Memristors". In: Advanced Functional Materials 32.15 (2022), p. 2111242. ISSN: 1616-3028. DOI: 10.1002/adfm.202111242.

- [56] Zhongrui Wang et al. “Memristors with Diffusive Dynamics as Synaptic Emulators for Neuro-morphic Computing”. In: Nature Mater 16.1 (Jan. 2017), pp. 101–108. ISSN: 1476-4660. DOI: 10.1038/nmat4756.
- [57] John Guckenheimer et al. “Bifurcation, Bursting, and Spike Frequency Adaptation”. In: ().
- [58] Yunus Babacan, Fırat Kaçar, and Koray Gürkan. “A Spiking and Bursting Neuron Circuit Based on Memristor”. In: Neurocomputing 203 (Aug. 26, 2016), pp. 86–91. ISSN: 0925-2312. DOI: 10.1016/j.neucom.2016.03.060.
- [59] James Kunert et al. The Role of Multistability and Transient Trajectories in Networked Dynamical Systems. Sept. 3, 2015. DOI: 10.48550/arXiv.1509.01224. arXiv: 1509.01224 [q-bio]. Pre-published.
- [60] J. P. Newman and R. J. Butera. “Mechanism, Dynamics, and Biological Existence of Multistability in a Large Class of Bursting Neurons”. In: Journal of Applied Physics 64.7 (Oct. 1, 1988), pp. 3620–3628. ISSN: 0021-8979, 1089-7550. DOI: 10.1063/1.341399. arXiv: 0810.1544 [q-bio].
- [61] Hairong Lin et al. “Review on Chaotic Dynamics of Memristive Neuron and Neural Network”. In: Nonlinear Dyn 106.1 (Sept. 1, 2021), pp. 959–973. ISSN: 1573-269X. DOI: 10.1007/s11071-021-06853-x.
- [62] Rodolfo R. Llinás and Mircea Steriade. “Bursting of Thalamic Neurons and States of Vigilance”. In: Journal of Neurophysiology 95.6 (June 2006), pp. 3297–3308. ISSN: 0022-3077. DOI: 10.1152/jn.00166.2006.
- [63] Rebecca A. Mease et al. “Multiplexed Spike Coding and Adaptation in the Thalamus”. In: Cell Reports 19.6 (May 9, 2017), pp. 1130–1140. ISSN: 2211-1247. DOI: 10.1016/j.celrep.2017.04.050.
- [64] Richard Naud and Henning Sprekeler. “Sparse Bursts Optimize Information Transmission in a Multiplexed Neural Code”. In: Proceedings of the National Academy of Sciences 115.27 (July 3, 2018), E6329–E6338. DOI: 10.1073/pnas.1720995115.
- [65] Eugene M. Izhikevich et al. “Bursts as a Unit of Neural Information: Selective Communication via Resonance”. In: Trends in Neurosciences 26.3 (Mar. 1, 2003), pp. 161–167. ISSN: 0166-2236, 1878-108X. DOI: 10.1016/S0166-2236(03)00034-1. PMID: 12591219.
- [66] Ambarish S. Pawar et al. “Mechanisms of Spatiotemporal Selectivity in Cortical Area MT”. In: Neuron 101.3 (Feb. 6, 2019), 514–527.e2. ISSN: 0896-6273. DOI: 10.1016/j.neuron.2018.12.002. PMID: 30606614.
- [67] Wenjia Meng et al. “Off-Policy Proximal Policy Optimization”. In: Proceedings of the AAAI Conference on Artificial Intelligence 37.8 (June 26, 2023), pp. 9162–9170. ISSN: 2374-3468. DOI: 10.1609/aaai.v37i8.26099.
- [68] Tianci Gao et al. Enhancing Sample Efficiency and Exploration in Reinforcement Learning through the Implicit Representation. Dec. 14, 2025. DOI: 10.48550/arXiv.2409.01427. arXiv: 2409.01427 [cs]. Pre-published.
- [69] Shashank Misra et al. “Probabilistic Neural Computing with Stochastic Devices”. In: Advanced Materials 35.37 (2023), p. 2204569. ISSN: 1521-4095. DOI: 10.1002/adma.202204569.
- [70] Gehua Ma, Rui Yan, and Huajin Tang. “Exploiting Noise as a Resource for Computation and Learning in Spiking Neural Networks”. In: Patterns 4.10 (Oct. 13, 2023), p. 100831. ISSN: 2666-3899. DOI: 10.1016/j.patter.2023.100831.
- [71] Dominik Dold et al. “Stochasticity from Function — Why the Bayesian Brain May Need No Noise”. In: Neural Networks 119 (Nov. 1, 2019), pp. 200–213. ISSN: 0893-6080. DOI: 10.1016/j.neunet.2019.08.002.

Appendix

Statement of Contribution

This work was supervised by Professor Alexander Balanov and Professor Sergey Saveliev at the department of Physics, Loughborough University.

The stochastic and deterministic transneuron model equations were derived in prior published work by Midya et al. [a] and provided to the author as a starting point for the investigation. Furthermore, the fabricated device and experimental characterisation described in Midya et al. [a] were similarly not part of this project.

All numerical analysis was conducted independently by the author, including the continuation analysis and regime mapping, spike train statistical characterisation, synchronisation and selectivity measurement, surrogate MLP design and training, and PPO optimisation framework implementation. All figures, analysis scripts, and written content are the author's own work.

AI assistance disclosure: Google Gemini (Google) was used as a writing and coding consultant during this project. Its use was limited to prose refinement, Latex formatting assistance, and Python code generation. All scientific judgements, interpretations, and conclusions are solely the author's own.

[a] Midya, R., Pawar, A. S., Pattnaik, D. P., Mooshagian, E., Borisov, P., Albright, T. D., Snyder, L. H., Williams, R. S., Yang, J. J., Balanov, A. G., Gepshtein, S., & Savel'ev, S. E. (2025). Artificial transneurons emulate neuronal activity in different areas of brain cortex. *Nature Communications*, 16(1), 7289. <https://doi.org/10.1038/s41467-025-62151-9>.

ODE simulation parameters

Table A1: Deterministic transneuron model parameters used throughout this work.

Transneuron Model Parameters			
Parameter	Symbol	Value	Description
Thermal charge	q_T	1.0	Thermophoretic coupling coefficient
Cooling constant	κ	0.9	Thermal relaxation rate
Thermal capacitance	C_{th}	0.18	Energy to temperature conversion rate
DC current offset	I_0	0.0	External current bias
Tunnelling decay length	λ	0.12	Characteristic tunnelling length scale
Capacitive timescale	τ	30.0	Voltage relaxation timescale
AC driving amplitude	V_{AC}	1.0	Harmonic forcing amplitude
Integration step size	h	1×10^{-3}	RK4 fixed time step

Note. All parameters are given in normalised units. The integration step size h was reduced to 5×10^{-4} in regions of fast dynamics.

Table A2: Results of Synchronisation Simulation Optimisation

Fixed Simulation Parameters: $V_{DC} = 6.0 \text{ V}$, $V_{AC} = 1.0 \text{ V}$, $R_{ext} = 500 \Omega$, $\omega = 0.1 \pm 0.1 \text{ Hz}$, $\omega_{step} = 10^{-4} \text{ Hz}$						
Optimization Results						
# Transient	# Measured	Width (Hz)	Centre	Depth	Time (mins)	Threads
200	500	0.0068	0.00665	0.0144	50	22
100	250	0.0067	0.066	0.014	10	4
50	125	0.0067	0.067	0.014	5	4
25	60	0.0067	0.0666	0.0143	5	1
15	30	0.0068	0.0666	0.0144	3	1
5	15	0.0069	0.0665	0.15	2	1

Note. Rows 1–5 represent stable measurement parameters. Row 6 indicates the point at which simulation results began to deteriorate.

Table A3: Linear regression results for spiking frequency in regime I

Fit Parameter	Value	Error
m	0.0837	± 0.016
c	-0.0832	± 0.0354
χ^2	0.067	N/A
$\hat{\chi}^2$	0.0013	N/A

Surrogate Model

Table A4: Architectural Specifications of the Surrogate MLP.

Architectural Component	Implementation	Details
Input Layer	2-node Input Vector	Inputs: $\{V_{ext}, R_{ext}\}$
Hidden Layers	1 Dense Layer (64 Units)	Tanh Activation Function
Output Layer	5-node Output Vector	Output Features $[w, CV_1, CV_2, f_c, d]$
Activation Logic	Output-Layer ReLU	Enforces non-negativity ($y \geq 0$)
Normalisation	Min-Max Scaling	Output interval $[0, 1]$
Optimisation	Adam Optimizer	MSE Loss minimization
Sampling Strategy	Latin Hypercube (LHS)	Parameter space coverage

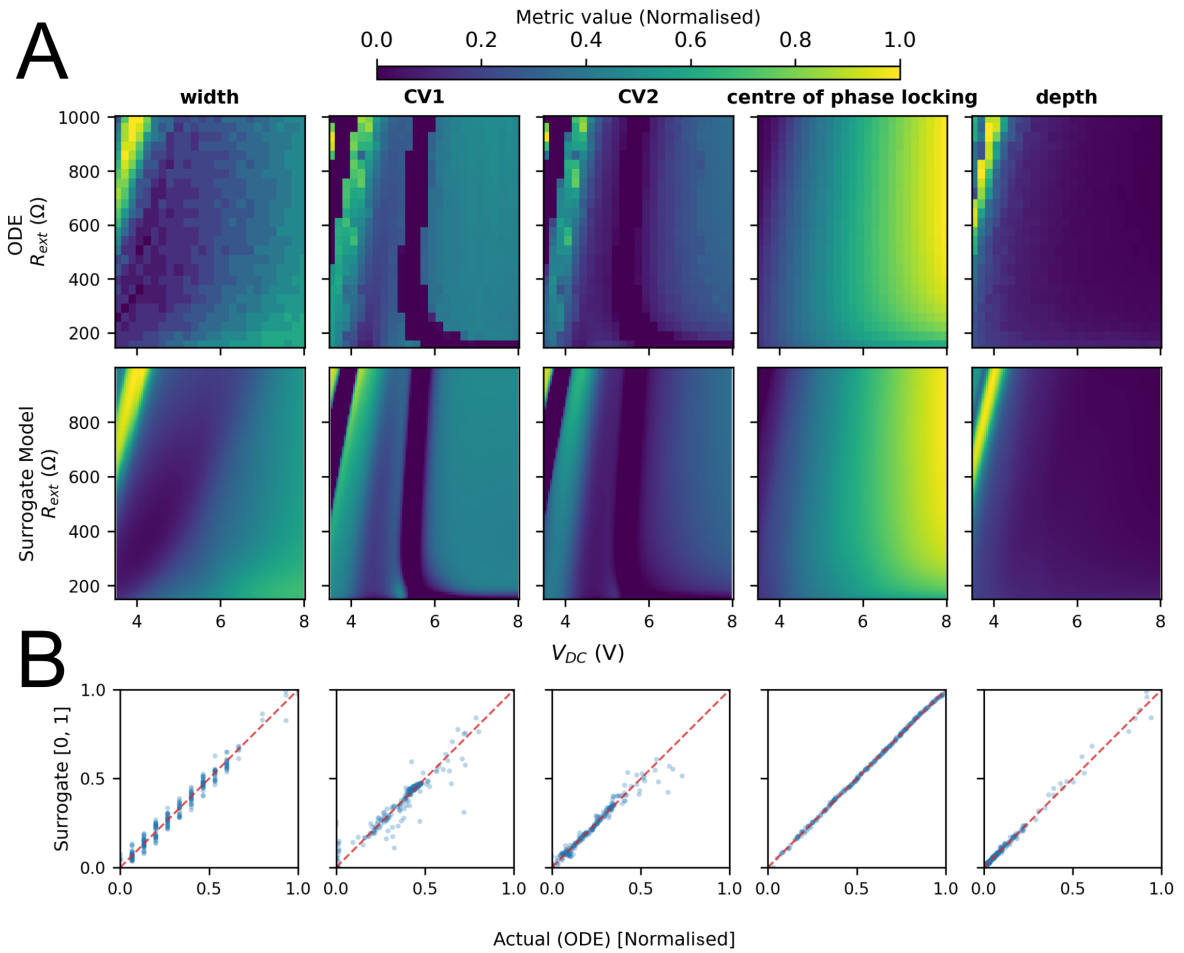


Figure A1: **Parametric heatmaps and plots comparing the original and surrogate models:** **A** The first row shows heatmaps generated using the standard ODE model, it took roughly 30 minutes to produce. The second row shows heatmaps based on predictions made by the surrogate model, which took seconds to produce. **B** Scatter plots of the actual ODE value vs the surrogate models production. The line $x=y$ indicates the line of no error, the more tightly the points are spread around this line, the better more accurate the surrogate model is.

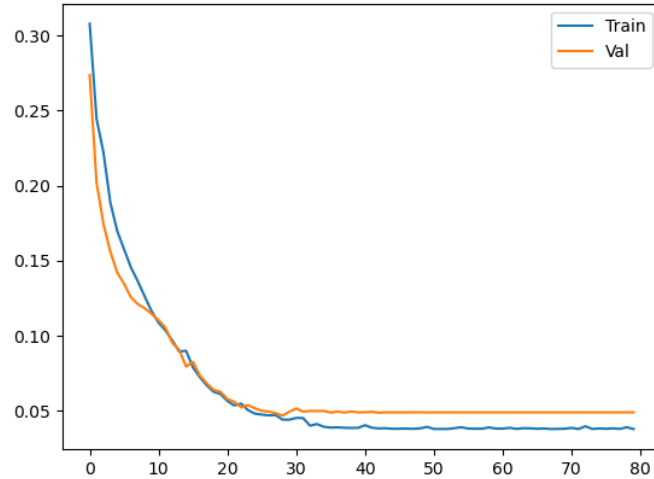


Figure A2: **Training loss for surrogate model:** Plot showing loss curve during training, evidencing convergence of the surrogate model.

PPO implementation

Table A5: PPO Configuration and Training Convergence Results

Category	Parameter / Metric	Specification / Final Value
Network	Policy Architecture	2-layer MLP (64 nodes/layer)
	Activation Function	Tanh
Learning	Base Learning Rate	5×10^{-5}
	Final Learning Rate	1.05×10^{-6}
	Mini-batch Size	64
	Total Training Steps	2,560,000
Objective	Clipping Range (ϵ)	0.2
	Entropy Coeff. (c_{ent})	0.01
	Final Entropy Loss	-1.66
Convergence Performance (at 500 Iterations)		
Rollout	Mean Reward	-123
	Mean Episode Length	100
Training	Explained Variance	0.997
	Approx. KL Divergence	2.745×10^{-5}
	Value Loss	0.71
	Policy Gradient Loss	-9.82×10^{-5}
Throughput	Frames Per Second (FPS)	1804
	Total Updates	9990